

PROGRAM:

```
#include<stdio.h>

#include<limits.h>

void insert(int ary[], int hFn, int size) {

    int element, pos, n = 0;

    printf("Enter key element to insert\n");

    scanf("%d", &element);

    pos = element % hFn;

    while (ary[pos] != INT_MIN) {

        // INT_MIN and INT_MAX indicate that cell is empty or deleted

        if (ary[pos] == INT_MAX)

            break;

        pos = (pos + 1) % hFn;

        n++;

        if (n == size) {

            printf("Hash table is full. No place to insert this element.\n");

            return;

        }

    }

    ary[pos] = element;

    // Inserting element

}

void delete(int ary[], int hFn, int size) {

    int element, pos, n = 0;

    printf("Enter element to delete\n");

    scanf("%d", &element);

    pos = element % hFn;

    while (n++ != size) {
```

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    if (ary[pos] == INT_MIN) {  
        printf("Element not found in hash table\n");  
        return;  
    }  
    else if (ary[pos] == element) {  
        ary[pos] = INT_MAX; // Mark as deleted  
        printf("Element deleted\n");  
        return;  
    }  
    else {  
        pos = (pos + 1) % hFn;  
    }  
}  
printf("Element not found in hash table\n");  
}  
  
void search(int ary[], int hFn, int size) {  
    int element, pos, n = 0;  
    printf("Enter element you want to search\n");  
    scanf("%d", &element);  
    pos = element % hFn;  
    while (n++ != size) {  
        if (ary[pos] == element) {  
            printf("Element found at index %d\n", pos);  
            return;  
        }  
        else if (ary[pos] == INT_MAX || ary[pos] != INT_MIN) {  
            pos = (pos + 1) % hFn;  
        }  
    }  
}
```

```

    }

    printf("Element not found in hash table\n");
}

void display(int ary[], int size) {
    printf("Index\tValue\n");
    for (int i = 0; i < size; i++) {
        printf("%d\t", i);

        if (ary[i] == INT_MIN) {
            printf("Empty\n");
        }

        else if (ary[i] == INT_MAX) {
            printf("Deleted\n");
        }

        else
        {
            printf("%d\n", ary[i]);
        }
    }
}

int main() {
    int size, hFn, choice;

    printf("Enter size of hash table\n");

    scanf("%d", &size);

    int ary[size];

    printf("Enter hash function (if mod 10, enter 10)\n");

    scanf("%d", &hFn);

    for (int i = 0; i < size; i++)
        ary[i] = INT_MIN;

```

```
// Assigning INT_MIN indicates that cell is empty

do {

    printf("\nEnter your choice\n");

    printf("1 -> Insert\n2 -> Delete\n3 -> Display\n4 -> Search\n0 -> Exit\n");

    scanf("%d", &choice);

    switch (choice) {

        case 1:

            insert(ary, hFn, size);

            break;

        case 2:

            delete(ary, hFn, size);

            break;

        case 3:

            display(ary, size);

            break;

        case 4:

            search(ary, hFn, size);

            break;

        case 0:

            printf("Exiting...\n");

            break;

        default:

            printf("Enter correct choice\n");

            break;

    }

} while (choice != 0);

return 0;

}
```

OUTPUT:

Enter size of hash table

5

Enter hash function (if mod 10, enter 10)

5

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

1

Enter key element to insert 12

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

1

Enter key element to insert 25

Enter your choice 1 -> Insert 2 -> Delete 3 -> Display 4 -> Search 0 -> Exit 1

Enter key element to insert 20

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

1

Enter key element to insert 8

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search 0 -> Exit

1

Enter key element to insert 18

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

3

Index Value

0 25

1 Empty

2 20

3 8

4 18

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

4

Enter element you want to search

20

Element found at index 2

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

2

Enter element to delete

8

Element deleted

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

3

Index Value

0 25

1 Empty

2 20

3 Deleted

4 18

Enter your choice

1 -> Insert

2 -> Delete

3 -> Display

4 -> Search

0 -> Exit

0

Exiting...